

MD ARIFUL HOQUE

PhD Candidate in Biostatistics & Data Analytics

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PROFESSIONAL SUMMARY

PhD Candidate in Biostatistics at Florida International University with methodological and applied research expertise in multicollinearity, regression modeling, statistical inference, survival analysis, simulation studies, epidemiology, and public health data analytics. Experienced in developing and evaluating regression and shrinkage estimators for correlated, high-dimensional, and complex health data, conducting Monte Carlo simulation studies, and analyzing large-scale epidemiologic and clinical datasets using advanced statistical methodologies. Emerging research interests extend to interpretable machine learning and statistically principled AI/LLM-enabled workflows, with emphasis on stability, evaluation, transparency, and responsible use. Interested in advanced undergraduate instruction and developing effective online learning materials. Collaborative researcher with interdisciplinary experience in environmental health, HIV/AIDS, COVID-19, oncology, and Alzheimer's disease research. Statistical researcher with strong experience in statistical programming, reproducible research, and methodological development, with additional hands-on training in clinical-trial statistical programming, CDISC SDTM/ADaM concepts, SAP implementation, and independent QC/validation.

RESEARCH INTERESTS

- Biostatistics and Public Health Data Analytics
- Regression Modeling and Statistical Inference
- Multicollinearity and Shrinkage Estimation
- High-dimensional Modeling
- Survival Analysis and Longitudinal Data Analysis
- Clinical Trial Design and Analysis
- Randomized Controlled Trial (RCT) Methodology
- Causal Inference and Mediation Analysis
- Indirect Treatment Comparison Methods
- Epidemiology and Population Health Research
- Environmental and Climate Health Statistics
- Multiple Testing Procedures
- Missing Data Methods and Multiple Imputation
- Monte Carlo Simulation Studies
- Machine Learning and Artificial Intelligence for Health Data
- Neural Networks and Deep Learning
- Robust Statistical Methods
- Computational and Reproducible Research

EDUCATION

Florida International University PhD Candidate (ABD) in Public Health Biostatistics and Data Analytics	Miami, FL, USA 2023 - Expected Summer 2027
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Florida International University MS in Statistics	Miami, FL, USA 2023
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University of Dhaka MS in Statistics	Dhaka, Bangladesh 2020
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University of Dhaka Bachelor of Science (Hons.) in Statistics	Dhaka, Bangladesh 2018
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RESEARCH EXPERIENCE

Graduate Assistant Department of Biostatistics	Florida International University August 2023 - Present
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- Conducted empirical **Type I error and statistical power** comparisons across multiple regression frameworks.
- Developed and evaluated one- and two-parameter regression estimators under **multicollinearity** using Monte Carlo simulation studies.
- Worked on methodological research involving **ridge regression, Liu estimators, Elastic Net, and robust** estimation techniques.
- Applied advanced regression methodologies to environmental and public health datasets using **R and SAS**.
- Contributed to **manuscript preparation, statistical reporting, and reproducible research** workflows.

Research Assistant
Department of Biostatistics

Florida International University
August 2023 - Present

- Working as a Research Assistant under **NIH RCMI** grant at Florida International University.
- Conducted statistical analyses for interdisciplinary public health and clinical research projects involving **maternal health, oncology, HIV/AIDS, COVID-19, climate health, and Alzheimer's disease**.
- Evaluated associations between **environmental metal mixtures and maternal/neonatal** outcomes applying advanced epidemiologic methods including logistic regression, Weighted Quantile Sum (WQS) regression, Bayesian Kernel Machine Regression (BKMR), and Structural Equation Modeling (SEM).
- Analyzed large-scale epidemiologic and clinical datasets to investigate **gestational hypertension, gestational diabetes, neonatal respiratory distress** syndrome, and related health outcomes.
- Conducted univariate and multivariable logistic regression analyses to evaluate predictors and barriers associated with fertility preservation consultation and egg-freezing completion among diverse **oncology** patients at an NCI-designated cancer center.
- Developed **survival and machine-learning** models using ADNI longitudinal **Alzheimer's disease** data to predict progression from cognitively normal (CN) to mild cognitive impairment (MCI) and dementia.
- Applied **Cox proportional hazards models, multi-state survival analysis, LASSO regression, and DeepSurv neural-network** modeling for longitudinal risk prediction and biomarker evaluation.
- Working on the **Climate Change and Community Health Consequences in Colombia (5C)** project involving population-based cohort studies on climate and health outcomes in South America.
- Conducted epidemiological analyses related to **HIV/AIDS, COVID-19**, and population health using large-scale health datasets.
- Volunteer research assistant on a systematic review project examining trends in biostatistical methods applied to **orthopedic cancer and joinpoint analysis** for Baptist Health.
- Supported literature **screening, data extraction, and methodological** review for collaborative research projects.
- Applied **Farrington Algorithm** to estimate the excess mortality rates among people with HIV in Florida during the early COVID-19 period in Florida.
- Compared the $\tilde{S}$ infection rates among sexual minority men in South Florida, both with and without HIV, using data from the **CRUSH (COVID-19 Research for Understanding the role of Substance use and HIV)** study.
- Managed and analyzed complex clinical and public health datasets, including **data harmonization, variable recoding, missing-data diagnostics, multiple imputation, sensitivity analyses, statistical visualization, and interpretation** according to epidemiologic best practices.
- Developed reproducible analytical workflows, publication-ready tables, forest plots, and statistical reports using **R, Quarto, and data visualization** packages while collaborating with interdisciplinary investigators on study design, analysis, and dissemination of research findings.

Graduate Assistant
Department of Statistics

Florida International University
August 2021 - July 2023

- Comparison between some one and two Parameter estimators for the **multicollinear gaussian linear regression model** using simulations and applications

Undergraduate Research Assistant
Department of Statistics

University of Dhaka
2017 - 2018

- Conducted research on the **socioeconomic impact of dengue** outbreaks in Dhaka city using survey-based statistical analysis.
- Contributed to data **collection, cleaning, visualization, and statistical analysis** for reports

TEACHING EXPERIENCE

Guest Instructor
Department of Biostatistics

Florida International University
May 2024 - August 2024

- Taught **Generalized Linear Models (Poisson), Ridge, LASSO, and Elastic Net** Regression
- Presented statistical concepts with applied modeling examples connecting theory, computation, and interpretation.

Teaching Assistant
Department of Biostatistics

Florida International University
August 2023 - Present

- Assisted Dr. Boubakari Ibrahimou in preparing and delivering statistics and data analysis for **R for Health Data Science** course materials.
- Conducted discussion sections, office hours, and academic support sessions for **Biostatistics I** course under Dr. Changwon Yoo.
- Supported **Applied Statistical Methods for Discrete Data** under Dr. Boubakari Ibrahimou, grading, assignment evaluation, and student mentoring in applied biostatistics coursework.

Graduate Teaching Assistant
Department of Statistics

Florida International University
August 2021 - July 2023

- Conducted discussion sections, proctored examinations, and provided valuable academic support for **undergraduate and graduate statistics courses**.

THESIS AND DISSERTATION

Florida International University

Department of Biostatistics

August 2023 - Present

- Test Statistics for **Testing** the Regression Coefficients for the One and Two Parameter **Multicollinear** Regression Models: An Empirical Comparison
- **Advisor:** Dr. Zoran Bursac and Dr. B. M. Golam Kibria

Florida International University

Department of Statistics

August 2021 - July 2023

- On Some Two parameters Biased Estimators for the **Logistic Regression Model**: Simulations and Applications
- **Advisor:** Dr. B. M. Golam Kibria

University of Dhaka

Department of Statistics

2019 - 2020

- Determining the Optimal **Changepoint Models** and Their Applications on Time Series Data.
- **Advisor:** Dr. Anamul Haque Sajib

PUBLICATIONS

Manuscript

- **Hoque, M. A.**, Bursac, Z., & Kibria, B. G. (2026). Performance of Some Test Statistics for Testing the Regression Coefficients for the One-and Two-Parameter Multicollinear Gaussian Multiple Linear Regression Models: An Empirical Comparison. In Advances in Shrinkage and Penalized Estimation Strategies: Honoring the Contributions of Professor AK Md. Ehsanes Saleh (pp. 327-357). Cham: *Springer Nature Switzerland*.
- **Hoque, M. A.**, Kibria, B. M. G., & Bursac, Z. (2026). New Two-Parameter Ridge Estimators for Addressing Multicollinearity in Linear Regression: Theory, Simulation, and Applications. *Stats*, 9(1), 15. <https://doi.org/10.3390/stats9010015>
- Dhungel, S., Ward, M.K., Barreto, G.A. et al. Excess Mortality Among People with HIV in Florida During the Initial Onset of the COVID-19 Pandemic: A Surveillance-Based Analysis. *textit{AIDS and Behavior}* (2025). <https://doi.org/10.1007/s10461-025-05014-0>
- **Hoque, M. A.**, Bursac, Z., & Kibria, B. M. G. (2025). Inferences About Two-Parameter Multicollinear Gaussian Linear Regression Models: An Empirical Type I Error and Power Comparison. *Stats*, 8(2), 28. <https://doi.org/10.3390/stats8020028>
- **Hoque, M. A.**, & Kibria, B. M. G. (2024). Performance of Some Estimators for the Multicollinear Logistic Regression Model: Theory, Simulation, and Applications. *Research in Statistics*, 2(1), 2364747.
- **Hoque, M. A.**, & Kibria, B. M. G. (2023). Some One and Two parameter Estimators for the Multicollinear Gaussian Linear Regression Model: Simulations and Applications. *Surveys in Mathematics and its Applications*, 18, 183-221.

Submitted/In Revision

- **Hoque M. A.**, Bursac Z., Kibria B. M. G. (2025). Addressing Multicollinearity Using Two-Parameter Estimators for the Linear Regression Model: Theory, Simulation and Application.

Conference Presentations

- Marbin S. J, Ganatra K., Escobar B., Han S., **Hoque, M. A.**, Patrizio, P. (Oct, 2026). BREAKING BARRIERS TO FERTILITY PRESERVATION (FP): A COMPREHENSIVE STUDY OF DIVERSE CANCER PATIENTS. *American Society for Reproductive Medicine - 2026*
- **Hoque M. A.**, Kibria B. M. G., Bursac Z. (March, 2026). New Two-Parameter Ridge Estimators for Improving the Linear Regression Model under Multicollinearity: Theory, Simulation, and Application. *2026 Annual ASA Florida Chapter Meeting*
- **Hoque, M. A.**, Bursac, Z., Kibria, B. M. G. (Aug, 2025). Testing the Coefficients for the Two-Parameter Multicollinear Linear Regression Model. *Joint Statistical Meetings – Speed Presentation-2025*
- **Hoque, M. A.**, & Kibria, B. M. G. (2023). Some One and Two Parameter Estimators for the Multicollinear Logistic Regression Model: Theory, Simulation, and Applications. *EIGHTH INTERNATIONAL WEBINAR ON RECENT TRENDS IN STATISTICAL THEORY AND APPLICATIONS-2023*.
- **Hoque, M. A.**, & Kibria, B. M. G. (2023). Some One and Two Parameter Estimators for the Multicollinear Logistic Regression Model: Theory, Simulation, and Applications. *Annual Mini-Conference in Statistical Methods and Mentoring*
- **Hoque M. A.**, Bursac Z., Kibria B. M. G. (March, 2026). New Two-Parameter Ridge Estimators for Improving the Linear Regression Model under Multicollinearity: Theory, Simulation, and Application. *Stempel College Research Day-2026*
- **Hoque M. A.**, Bursac Z., Kibria B. M. G. (Aug, 2025). On Some Test Statistics for Testing the Regression Coefficients for the Two Parameter Multicollinear Regression Models: Simulation and Applications. *Stempel College Research Day-2025*

Selected Research Projects

- **Hoque, M. A., & Bursac, Z. (2023).** Regression Analysis Using IML (Interactive Matrix Language).
- **Hoque, M. A., Cao, L., Kammili, J. L., Vazquez, R., Yasmin, N. & Chen, Z. (2023).** Machine Learning for Breast Cancer Diagnosis.
- **Hoque, M. A., Yasmin, N., Islam, T. & George, F. (2022).** Hurricane Harvey: Utility Disruption, Damage, and Household Wellbeing.
- **Hoque, M. A., & Gulati, S. (2022).** Nonparametric Regression Analysis on Portland Cement Data.
- **Hoque, M. A., Shylouskaya, K., Roby, M. & Wu, W. (2022).** Prediction of Algerian Fires in the Bejaia and Sidi-Belle Region.

Honors and Grants

Graduate Assistantship

Florida International University

2023

- Scholarship support for Doctoral studies

Graduate Teaching Assistantship

Florida International University

2021

- Scholarship to conduct Graduate studies

Bangladesh Sweden Trust Fund Scholarship

Ministry of Finance, Bangladesh

2021

- Bangladeshi students enrolled at foreign university

Government Honors Scholarship

University of Dhaka

2018

- Based on merit

Professional Memberships

- Member, American Statistical Association (ASA)

SKILLS

- **Research Expertise** - Regression Modeling, Survival Analysis, Longitudinal Data Analysis, Clinical Trial Design, Randomized Controlled Trial (RCT) Methodology, Missing Data Analysis (MCAR, MAR, MNAR), Multiple Imputation (MICE), Sensitivity Analysis, Causal Inference, Mediation Analysis, Indirect Treatment Comparison Methods (MAIC, STC), Shrinkage Estimation, Machine Learning, Neural Networks, Deep Learning, Large Language Models, Explainable AI, Epidemiologic Methods
- **Software** - Excel, SPSS, STATA
- **Programming Languages** - R (tidyverse, dplyr, ggplot2, survival, glmnet, broom, surveillance), SAS (PROC LOGISTIC, GENMOD, PHREG, MIXED, SAS ODS), SQL, Python (pandas, NumPy, SciPy, statsmodels, scikit-learn), SPSS, STATA, Excel.
- **Machine Learning & AI:** Scikit-learn, PyTorch, HuggingFace Transformers, Neural Networks, Deep Learning (DeepSurv), Predictive Modeling; familiarity with modern AI/LLMs, and LangChain for LLM-based workflows
- **Visualization:** ggplot2, Seaborn, Matplotlib, Tableau, Power BI
- **Technical Publication** - Quarto, Markdown/Rmarkdown, LaTeX
- **DevOps** - Git, GitHub, REDCap, HPC/Slurm
- **Languages** - Bangla (Native), English (Professional Proficiency)

PROFESSIONAL DEVELOPMENT & CERTIFICATES

- **SQL, PowerBI, Tableau Essential Training** — LinkedIn Learning, 2026
- **Data Analysis with Python and Pandas** — LinkedIn Learning, 2026
- **Machine Learning with scikit-learn** — LinkedIn Learning, 2026
- Expanding a reproducible GitHub portfolio in **Python, SQL, and machine learning** for health-data analytics, including data preparation, **predictive modeling**, model evaluation, and transparent computational workflows. **LLM/RAG** extensions will be added after implementation and validation.